

(A). $\frac{-1}{2(2x+3)} + C$ (B) $\frac{-1}{(2x+3)} + C$ (C) $\frac{1}{(x+3)} + C$ (D) $\frac{-2}{(x+3)} + C$

QUESTION 1 (11 marks)

(a) A box has 4 Geography books and 3 Mathematics books in it.
Two books are selected at random from the box.

i. Draw a tree diagram to show all the possible outcomes. 2

ii. Find the probability that:

α . the two books are Mathematics books. 2

β . at least one of the books is a Geography book 1

(b) In a group of 80 students, 65 study maths, 45 study science and 5 do not study either maths or science. What is the probability that the student:

(i) studies maths and science. 1

(ii) studies maths or science. 1

(iii) studies only one of the two subjects. 2

(iv) does not study science. 2

QUESTION 2 (12 marks)

Consider the curve given by the equation $y = 9x(x - 2)^2$.

i) Find the co-ordinates of the stationary points and determine their nature. 5

ii) Find the co-ordinates of any points of inflexion. 2

iii) Sketch the curve in the domain $-1 \leq x \leq 3$. 3

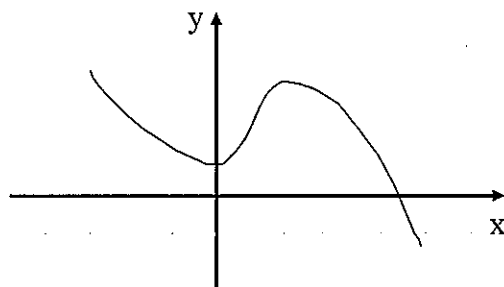
iv) What is the maximum value of $9x(x - 2)^2$ in the domain $-1 \leq x \leq 3$ 2

Question 3. (9 marks)

- (a) Find the range of values of x for which the curve
 $y = 2x^3 - 3x^2 - 12x + 8$ is concave up

2

- (b) The diagram shows the graph of a certain function $f(x)$.



Copy this graph into your writing booklet. On the same set of axes, draw a sketch of the derivative $f'(x)$ of the function.

4

- (c) Prove that $f(x) = 2x^3 - 3x^2 + 5x + 1$ has no stationary points.

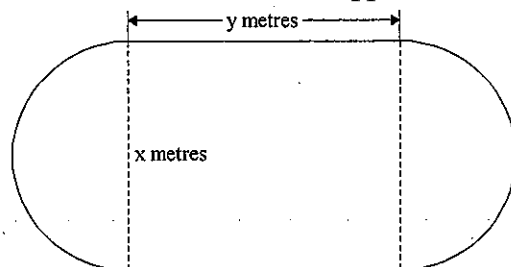
2

QUESTION 4 (11marks)

- (a) Find the value of k such that $\int_1^k \frac{4}{x^2} dx = 3$

3

- (b) A railway enthusiast designs a miniature railway of length 1000 metres.
 The route consists of two semicircles at opposite ends of a rectangle.



- i.) If the rectangle has a length of y metres and its width is x metres, show that:

$$y = 500 - \frac{\pi x}{2}.$$

2

- ii.) Show that the area, A , enclosed by the railway track is

given by
$$A = \frac{2000x - \pi x^2}{4}.$$

3

- iii.) Find the maximum area, to the nearest hectare, enclosed by the railway track.

3

Question 5.(23 marks)

(a) Find each indefinite integral:

(i) $\int \frac{dx}{x^5}$ 1

(ii) $\int x^2(5-3x)dx$ 2

(iii) $\int \frac{dx}{\sqrt{9-2x}}$ 3

(iv) $\int (3x+1)^4 dx$ 2

(v) $\int \frac{2x^3 - x^4}{4x} dx$ 3

(b) Evaluate:

(i) $\int_1^9 (1 + \sqrt{x}) dx$ 3

(ii) $\int_2^3 (2x-5)^3 dx$ 3

(iii) $\int_0^1 \frac{x^3 - 2x^2 + 3x}{x} dx$ 3

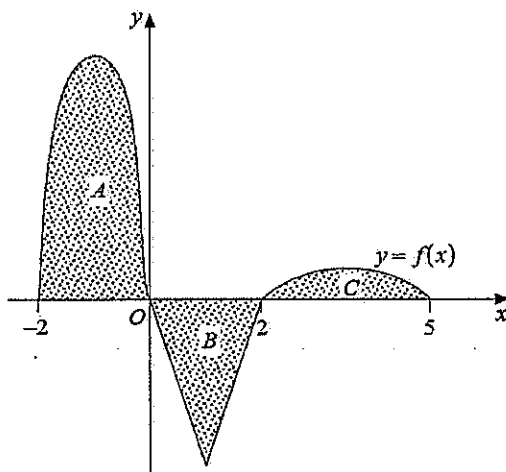
(iv) $\int_1^2 \frac{dx}{(3x-1)^2}$ 3

Question 6.(11 marks)

(a) The curve $y = f(x)$ has gradient function $\frac{dy}{dx} = 3x^2 - 2x + 1$. The curve passes through the point $Q(2, 3)$. Find its equation.

3

(b)

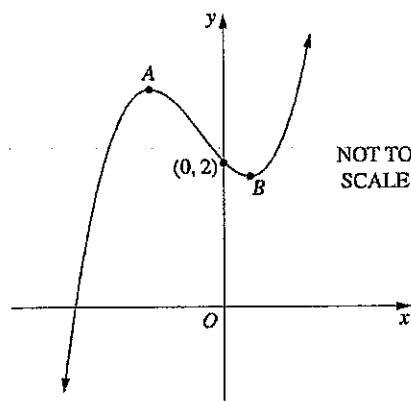


The graph of the function f is shown in the diagram. The shaded areas are bounded by $y = f(x)$ and the x axis. The shaded area A is 8 square units, the shaded area B is 3 square units and the shaded area C is 1 square unit.

Evaluate $\int_{-2}^5 f(x) dx$.

3

(b) The graph of $y = x^3 + x^2 - x + 2$ is sketched below. The points A and B are the turning points.



(i) Find the coordinates of A and B .

3

(ii) For what values of x is the curve concave up? Give reasons for your answer.

2

End of Test

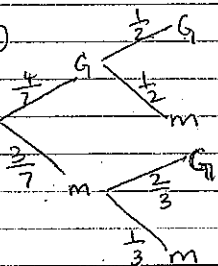
Part A

- 1) B
2) A
3) D
4) C
5) A

Part B

Question 1

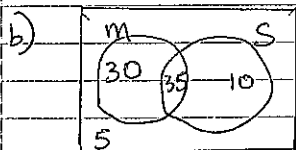
a) (i)



(2)

(ii) $\alpha) \frac{3}{7} \times \frac{1}{3} = \frac{1}{7}$ (2)

(iii) $\beta) 1 - \frac{1}{7} = \frac{6}{7}$ (1)



(i) $\frac{35}{80} = \frac{7}{16}$ (1)

(ii) $\frac{75}{80} = \frac{15}{16}$ (1)

(iii) $\frac{40}{80} = \frac{1}{2}$ (2)

(iv) $\frac{35}{80} = \frac{7}{16}$ (2)

Question 2

(i) $y = 9x(x-2)^2$

$$\frac{dy}{dx} = 9[2x(x-2) + (x-2)^2]$$

$$= 9(x-2)[2x + x-2]$$

$$= 9(x-2)(3x-2) \quad (2)$$

For stationary points,

$$\frac{dy}{dx} = 0 \Rightarrow 9(x-2)(3x-2) = 0$$

$$\therefore x = 2, \frac{2}{3} \quad (1)$$

When $x = 2$, $y = 9 \cdot 2 \cdot 0 = 0$

when $x = \frac{2}{3}$, $y = 9 \cdot \frac{2}{3} \cdot \frac{16}{9}$

$$= 10\frac{2}{3} \quad (2)$$

$\therefore$ stationary points are

$(2, 0)$ and $(\frac{2}{3}, 10\frac{2}{3})$ (2)

(ii) $\frac{dy}{dx} = 9(x^2 - 8x + 4)$

For possible points of inflexion, $\frac{d^2y}{dx^2} = 0$

$$\frac{d^2y}{dx^2} = 6x - 8 = 0 \Rightarrow x = \frac{4}{3}$$

Concavity test.

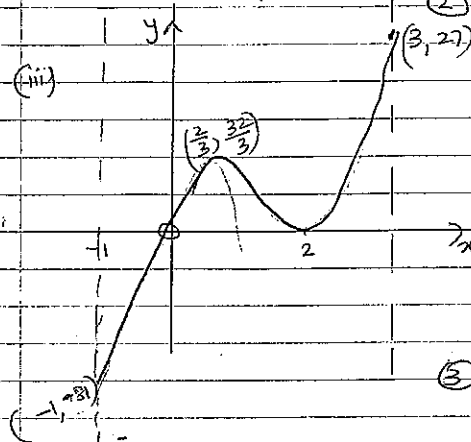
x	1	$\frac{4}{3}$	2
y''	-2	0	4

$\therefore$ Concavity changes when $x = \frac{4}{3}$

when $x = \frac{4}{3}$, $y = \frac{16}{3}$

$\therefore$ point of inflexion at

$(\frac{4}{3}, \frac{16}{3})$ (2)



(iv) when $x = 3$, $y = 27$

$\therefore$ maximum value = 27 (2)

Question 3

a) $y = 2x^3 - 3x^2 - 12x + 8$

$$y' = 6x^2 - 6x - 12$$

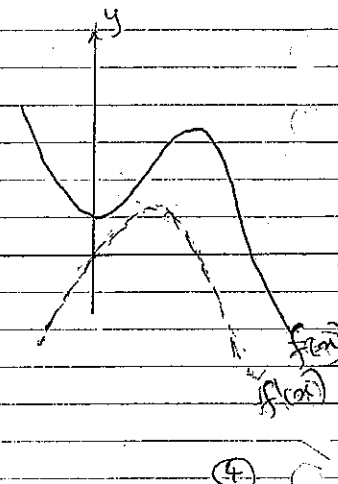
$$y'' = 12x - 6$$

concave up $\Rightarrow y'' > 0$

$$\therefore 12x - 6 > 0$$

$$\therefore x > \frac{1}{2} \quad (2)$$

b)



(c) $f(x) = 2x^3 - 3x^2 + 5x + 1$

$$f'(x) = 6x^2 - 6x + 5$$

For stationary points, $f'(x) = 0$

$$\therefore 6x^2 - 6x + 5 = 0$$

has no solutions as $\Delta < 0$

$\therefore$ has no stationary point (2)

Question 4

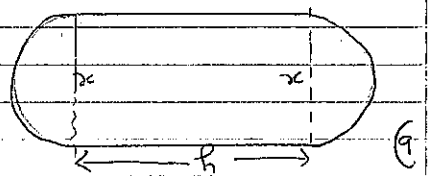
a) $\int_k^1 \frac{x^2}{4} dx = 3$

$\int_k^1 \frac{x^2}{4} dx = \int_k^1 \frac{1}{4} x^2 dx = \left[\frac{1}{12} x^3 \right]_k^1 = \frac{1}{12} (1 - k^3) = 3$

$\Rightarrow \frac{1}{12} (1 - k^3) = 3 \Rightarrow 1 - k^3 = 36 \Rightarrow k^3 = -35 \Rightarrow k = \sqrt[3]{-35}$

$\frac{1}{12} [1 - k^3] = 3 \Rightarrow 1 - k^3 = 36 \Rightarrow k^3 = -35 \Rightarrow k = \sqrt[3]{-35}$

$\frac{1}{12} [1 - k^3] = 3 \Rightarrow 1 - k^3 = 36 \Rightarrow k^3 = -35 \Rightarrow k = \sqrt[3]{-35}$



(1) $\pi x + 2y = 1000$

$y = \frac{1000 - \pi x}{2}$

$= 500 - \frac{\pi x}{2}$

(ii) Area $A = xy + \pi \frac{x^2}{2}$

$= x(500 - \frac{\pi x}{2}) + \pi \frac{x^2}{2}$

$= 500x - \frac{\pi x^2}{2} + \pi \frac{x^2}{2}$

$= 500x - \frac{\pi x^2}{2}$

(3)

(ii) $\frac{dA}{dx} = \frac{1}{4} (2000 - 2\pi x)$

Maximum area $\Rightarrow \frac{dA}{dx} = 0$

$\frac{1}{4} (2000 - 2\pi x) = 0$

$\therefore 2\pi x = 2000$

$x = \frac{2000}{2\pi} = \frac{1000}{\pi}$

$\frac{d^2A}{dx^2} = -2\pi < 0$

Maximum area at $x = \frac{1000}{\pi}$

Max. Area $= \frac{1}{4} [2000 \cdot \frac{1000}{\pi} - \pi (\frac{1000}{\pi})^2]$

$= \frac{1}{4} [1000000 (\frac{2}{\pi} - 1)]$

$= \frac{1}{4} [1000000 (\frac{2}{\pi} - 1)] = 7.95 \times 10^4$ Hectares

(3)

Question 5

(i) $\int \frac{dx}{x^5} = \int x^{-5} dx = \frac{x^{-4}}{-4} + C = -\frac{1}{4x^4} + C$

$= -\frac{1}{4x^4} + C$

(1)

(ii) $\int x^2 (5 - 3x) dx = \int (5x^2 - 3x^3) dx$

$= 5 \cdot \frac{x^3}{3} - 3 \cdot \frac{x^4}{4} + C$

(2)

(iii) $\int \frac{dx}{\sqrt{9 - 2x}} = \int (9 - 2x)^{-\frac{1}{2}} dx$

$= 2(9 - 2x)^{\frac{1}{2}} + C$

(3)

$= -\sqrt{9 - 2x} + C$

(3)

(iv) $\int (3x+1)^4 dx$

$= \frac{5 \times 3}{(3x+1)^5} + C$

(3)

(v) $\int \frac{2x^3 - x^4}{4x^4} dx$

$= \frac{1}{4} \int \frac{2x^3 - x^4}{x^4} dx$

$= \frac{1}{4} \int \frac{2x^3}{x^4} - \frac{x^4}{x^4} dx$

(3)

b) (i) $\int_1^9 (1 + \sqrt{x}) dx$

$= \left[x + \frac{2}{3} x^{\frac{3}{2}} \right]_1^9$

$= (9 + \frac{2}{3} \cdot 9^{\frac{3}{2}}) - (1 + \frac{2}{3})$

$= 9 + 18 - \frac{5}{3} = \frac{3}{26}$

(3)

(ii) $\int_3^2 (2x-5)^4 dx$

$= \left[\frac{(2x-5)^5}{5} \right]_3^2$

$= \frac{1}{5} [(-1)^5 - (-4)^5] = 0$

(3)

(iii) $\int_1^0 \frac{dx}{x^2 - 2x^2 + 3x}$

$= \int_1^0 \frac{dx}{x^2 - 2x^2 + 3x}$

$= \int_1^0 \frac{dx}{x^2 - 2x^2 + 3x}$

$= \frac{1}{3} - 1 + 3 - 0 = \frac{3}{7}$

(3)

(iv) $\int_2^1 \frac{dx}{(3x-1)^2}$

$= \int_2^1 \frac{dx}{(3x-1)^2}$

$= -\frac{1}{3} \left(\frac{1}{3} - \frac{1}{3} \right) = \frac{10}{3}$

(3)

Question 6

a) $\frac{dy}{dx} = 3x^2 - 2x + 1$

$\therefore y = \int (3x^2 - 2x + 1) dx + c$

$= x^3 - x^2 + x + c$

passes through (2, 3)

$\Rightarrow 3 = 2^3 - 2^2 + 2 + c$

$\therefore c = -3$

$\therefore y = x^3 - x^2 + x - 3$

(3)

b) $\int_{-2}^5 f(x) dx = \text{Area A} - \text{Area B} + \text{Area C}$
 $= 8 - 3 + 1 = 6 \text{ u}^2$

c) (i) $y = x^3 + x^2 - x + 2$

$\frac{dy}{dx} = 3x^2 + 2x - 1$

$= (3x - 1)(x + 1)$

Turning points occur when

$\frac{dy}{dx} = 0$

$\therefore (3x - 1)(x + 1) = 0$

$\therefore x = \frac{1}{3} \text{ or } -1$

when $x = \frac{1}{3}$, $y = \frac{49}{27}$

when $x = -1$, $y = 3$

$\therefore$ co-ordinates of A = (-1, 3)

B = $(\frac{1}{3}, \frac{49}{27})$

(ii) $\frac{d^2y}{dx^2} = 6x + 2$

Concave up $\Rightarrow \frac{d^2y}{dx^2} > 0$

$\therefore 6x + 2 > 0$

$\therefore x > -\frac{1}{3}$